

ALESSIO BRINI, PH.D.

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RESEARCH INTERESTS

Machine learning and reinforcement learning for financial decision-making, with applications to dynamic trading and derivative hedging. Financial econometrics and forecasting, with graph-based and time series foundation models for volatility and mixed-frequency models for macroeconomic data. Digital assets and decentralized finance, in particular liquidity provision on decentralized exchanges and anti-money-laundering analytics on blockchain data.

EDUCATION

Ph.D. at Scuola Normale Superiore 11/2018 - 09/2022
Mathematical Finance. *Pisa, Italy*

- Thesis: *Reinforcement learning for sequential decision-making: a data-driven approach for finance*

Master's degree at Università degli studi di Firenze 2016 - 2018
Quantitative Finance and Risk Management *Florence, Italy*

- Thesis: "Beyond the parameters' estimation burden: a Bayesian framework for option pricing."

Bachelor's degree at Università degli studi di Firenze 2012 - 2015
Business and Economics. *Florence, Italy*

ACADEMIC APPOINTMENTS

Executive in Residence - Duke University - Pratt School of Engineering
07/01/2024 - Ongoing, *Durham, NC, USA*

- Leads the *Digital Asset Research & Engineering Collaborative (DAREC)* Lab.
- Serves as a faculty member in FinTech and applies machine learning and data science methodologies to financial challenges, particularly in digital assets and DeFi, while delivering advanced coursework and leading research projects in these areas.

Postdoctoral Researcher - Duke University - Pratt School of Engineering
01/23/2022 - 06/30/2024, *Durham, NC, USA*

- Applied machine learning and data science methodologies to address financial challenges, specializing in digital assets and DeFi. Developed proficiency in diverse machine learning paradigms (supervised, unsupervised, reinforcement learning).
- Delivered coursework on *Machine Learning for FinTech*, *Data Wrangling and Visualization with Python*, and *Introduction to Statistics* for the Master of Engineering in Financial Technology program.
- Directed FinTech Capstone Projects spanning Machine Learning, Data Science, and Digital Ledger Technologies.
- Led summer research projects on DeFi and machine learning for graduate students.

PUBLICATIONS

Peer-Reviewed Publications

1. “On Deep Reinforcement Learning for Dynamic Trading with PPO: Challenges and Future Directions” (with Petter Kolm). *The Journal of Financial Data Science*. 2026
2. “Beyond Average Hive Performance: Tail Risk Measurement in Italian Apiculture with Honey-at-Risk” (with Giacomo Toscano, Ginevra Virginia Lombardi, and Maria Elvira Mancino). *Environmetrics*. 2026
3. “StableAML: Machine Learning for Behavioral Wallet Detection in Stablecoin Anti-Money Laundering on Ethereum” (with Luciano Juvinski and Haochen Li). *Blockchain*. 2026
4. “Graph Neural Networks for Bridge Swap Link Prediction in Uniswap v3” (with Qingran Zhou and Eric Liu). *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF’25)*. 2025
5. “SpotV2Net: Multivariate Intraday Spot Volatility Forecasting via Vol-of-Vol-Informed Graph Attention Networks” (with Giacomo Toscano). *International Journal of Forecasting*. 2025
6. “Modeling and Simulating a Startup Ecosystem Development with a Delayed Differential Equation” (with Viviana Fanelli). *Soft Computing*. 2025
7. “Data-driven Derivative Hedging with Quadratic Variation Penalty” (with Giacomo Domeniconi and Ali Fathi). *Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF’24)*. 2024
8. “Pricing Cryptocurrency Options with Machine Learning Regression for Handling Market Volatility” (with Jimmie Lenz). *Economic Modelling*. 2024
9. “A Comparison of Cryptocurrency Volatility-benchmarking New and Mature Asset Classes” (with Jimmie Lenz). *Financial Innovation*. 2024
10. “Reinforcement Learning Policy Recommendation for Interbank Network Stability” (with Gabriele Tedeschi and Daniele Tantari). *Journal of Financial Stability*. 2023
11. “Deep Reinforcement Trading with Predictable Returns” (with Daniele Tantari). *Physica A: Statistical Mechanics and its Applications*. 2023
12. “Assessing the resiliency of investors against cryptocurrency market crashes through the leverage effect” (with Jimmie Lenz). *Economics Letters*. 2022

Working Papers

1. “A Nonlinear Target-Factor Model with Attention Mechanism for Mixed-Frequency Data” (with Ekaterina Seregina). [arXiv:2601.16274](https://arxiv.org/abs/2601.16274). 2026
2. “Forecasting Realized Volatility with Time Series Foundation Models: A Comparison with Econometric Benchmarks”. [arXiv:2607.05291](https://arxiv.org/abs/2607.05291). 2026
3. “Empirical Models of the Time Evolution of SPX Option Prices” (with David Hsieh, Patrick Kuiper, Sean Moushegian, and David Ye). [arXiv:2506.17511](https://arxiv.org/abs/2506.17511). 2025

4. “Improving DeFi Accessibility through Efficient Liquidity Provisioning with Deep Reinforcement Learning” (with Haonan Xu). [arXiv:2501.07508](#). 2025
5. “A Machine Learning Approach to Forecasting Honey Production with Tree-Based Methods” (with Elisa Giovannini and Elia Smaniotto). [arXiv:2304.01215](#). 2023

Other Articles and Lightly Refereed Publications

- “Crypto dynamics during market downturns: Another dotcom boom-bust cycle?” (with Jimmie Lenz and Emma Rasiel). DAREC working paper. 2022
- “Bitcoin ETFs: Measuring the performance of this new market niche” (with Jimmie Lenz). [SSRN 4157711](#). 2022

GRANTS & FUNDING

- Co-Investigator on the project “Risk Management in Times of Unprecedented Geo-Political Volatility: A Machine-Learning Approach”, funded by the Institut Europlace de Finance (IEF), Institut Louis Bachelier (Grant amount: 10k €).
- Collaborator in the “BEEkeepers Weather Indexed INSurance” project at the University of Florence, funded by the Italian Ministry of Agricultural, Food and Forestry Policies (Grant amount: 600k €).

CONFERENCE PRESENTATIONS & INVITED TALKS

AIBThings 2026: IEEE 4th International Conference on Artificial Intelligence, Blockchain, and Internet of Things - Central Michigan University, Mount Pleasant, MI, September 5-6, 2026:

- Contributed Talk: *Do Rollups Self-Organize? Multi-Agent Reinforcement Learning in the EIP-4844 Blob Fee Market*.

Quaint Quant Conference - SMU Cox School of Business, Dallas, TX, April 10, 2026:

- Contributed Talk: *Attention-Based Mixed-Frequency Factor Models for Macroeconomic Forecasting*.

Conference on Society-Centered AI - Duke University, Durham, NC, February 12-14, 2026:

- Invited Talk (New Faculty Research Spotlight): *AI in Finance and Economics: From Data to Decisions that Matter*.

ICAIF’25: 6th ACM International Conference on AI in Finance - Singapore, November 16-17, 2025:

- Contributed Talk: *Graph Neural Networks for Bridge Swap Link Prediction in Uniswap v3*. Honorable mention for best paper award.

AI for Social Impact: Bridging Innovations in Finance, Social Media, and Crime Prevention - AAAI Conference - Philadelphia, March 3, 2025:

- Contributed Talk: *Improving DeFi Accessibility through Efficient Liquidity Provisioning with Deep Reinforcement Learning*. Best paper award ([link](#)).

ICAIF’24: 5th ACM International Conference on AI in Finance - New York, November 14-17, 2024:

- Poster Presentation: *Data-driven Derivative Hedging with Quadratic Variation Penalty*.

2024 NBER-NSF Time Series Conference - Philadelphia, September 20-21, 2024:

- Poster Presentation: *SpotV2Net: Multivariate intraday spot volatility forecasting via vol-of-vol-informed graph attention networks.*

Machine Learning of Dynamic Processes and Time Series Analysis - Pisa, November 26-27, 2020:

- Contributed Talk: *Trading Mean Reversion with Reinforcement Learning.*

13th European Financial Mathematics Summer School - Vienna, August 31 - September 4, 2020:

- Contributed Talk: *Trading Mean Reversion with Reinforcement Learning.*

Eastern European Machine Learning Summer School - Krakow, July 1-9, 2020:

- Poster Presentation: *Trading Mean Reversion with Reinforcement Learning.*

TEACHING EXPERIENCE

FINTECH 540: Machine Learning for FinTech - Pratt School of Engineering, Duke University:

Explored the impact of machine learning on the rapidly evolving FinTech industry through both theoretical understanding and practical application, with extensive coding sessions utilizing real financial data.

FINTECH/ECE 590: Data Wrangling and Visualization with Python - Pratt School of Engineering, Duke University:

Leveraged Python for data querying, cleaning, manipulation, and visualization. Covered advanced topics such as Databases (MySQL, NoSQL), Web Scraping, Data Manipulation, and Data Collection.

FINTECH 502: FinTech Capstones - Pratt School of Engineering, Duke University:

Oversaw teams of students developing real-world technical solutions in the FinTech industry, with projects sponsored by industry professionals.

FINTECH 520: Introduction to Statistics and Econometrics - Pratt School of Engineering, Duke University:

Designed specifically for graduate students in Financial Technology, this course established fundamental statistical concepts and tools essential for advanced FinTech studies and research.

Python for Data Science - University of Florence (Italy) - March-May, Sep-Dec, 2020:

Provided an introduction to Python, covering the basics of the language structure and the main packages of the scientific stack for data analysis and extraction of meaningful information.

ADVISING & MENTORING

Graduate Students:

- Duccio Guerrini, University of Florence, 2020: “Stock market indexes and machine learning algorithms. Replication of the S&P500 index by the autoencoder neural network.”
- Dmitry Guzairov, University of Florence, 2021: “Hierarchical risk-parity portfolios: a clustering approach to the investment problem”

- Niccolò Tarchi, University of Florence, 2021: “Transformer-based approach for creating financial portfolios”
- Mike Liu, Duke University, 2022: Independent Study on Asset Pricing and Financial Dashboard development
- Alex Ilgenfritz, Duke University, 2023: “Optimal Liquidity Provision in Uniswap v3 with Reinforcement Learning”
- Haonan Xu, Duke University, 2024: “Improving DeFi Accessibility through Efficient Liquidity Provisioning with Deep Reinforcement Learning”
- Qingran Zhou, Duke University, 2025: “Graph Neural Networks for Bridge Swap Link Detection on Uniswap”
- Longyi Hu, Duke University, 2025: “Conformal Prediction for Finance and Economics.”
- Luciano Juvinski, Duke University, 2025: “A Scalable Framework for Stablecoin AML: Leveraging Supervised Learning Models for Behavioral Wallet Detection on Ethereum”

PROFESSIONAL SERVICE

Editorial Service:

- Cogent Economics and Finance (Taylor & Francis), Associate Editor (2026-Ongoing)

Program Committee:

- ACM International Conference on AI in Finance (ICAIF) 2026, program committee member and meta-reviewer
- IEEE International Conference on Big Data (IEEE BigData) 2026, program committee member

Referee Service:

Journal of Financial Econometrics, Annals of Operations Research, Expert Systems with Applications, Financial Innovation, Finance Research Letters, Applied Economics Letters, Scientific Reports, Applied Economics, Journal of Applied Economics, Discover Artificial Intelligence, AI Computer Science and Robotics Technology, Data Science and Management, ACM International Conference on AI in Finance (ICAIF) 2024–2025, European Conference in AI for Finance (ECAIF) 2024–2025, IEEE International Conference on Machine Learning and Applications (ICMLA) 2026

Conference Organization:

- Scientific Committee - Florence-Paris Workshop: Statistics of Random Processes and Its Applications to Financial Econometrics, July 2023, Florence, Italy
- Organizer - Digital Assets at Duke (DA@D), January 2023 and 2024, Durham, NC, USA

Departmental Service (Duke University, Pratt School of Engineering):

- Assisting with recruitment and admissions review for the FinTech graduate program (2023-Ongoing).
- Enhancing program outreach within the industry and research landscapes.